E-solution Finance Service POD Technical Design

Version 1.1 delivery design | 29 October 2026 | Audience: client finance owners, implementation engineers and acceptance reviewers

1 Purpose and delivery decision

E-solution Finance Service POD provides five governed finance workflows over authorized QuickBooks Desktop or Enterprise read-only exports. It combines source-linked calculations, bounded AI drafting and extraction, individual user access, human approvals, external email and a manual accounting task register. Finance Admin posts accounting outside this application and supplies refreshed evidence for verification.

This document describes the implementation in the supplied source tree. It is a technical design and test design document, not an assertion that every client operating prerequisite has been completed. The deployment is suitable for controlled acceptance testing once the client supplies the required evidence and accounts. Unattended production service requires the operational gates in Section 15. The final release review and fresh test run are recorded separately; old test counts and old deployment snapshots must not be treated as current acceptance evidence.

The requirements baseline is the 32 physical slides in `Finance and Accounting AI agentsV1.1.pptx`. Printed slide labels skip numbers; references below use physical positions. Slides 1-2 define the product and roles, 3-8 AR, 9-12 bank reconciliation, 13-20 FP&A, 21-28 treasury, 29-31 reporting, and 32 shared acceptance controls. Earlier automatic QuickBooks invoice-note or payment-overlay designs are superseded by the repeated read-only policy.

Problem statement

Finance teams need to move from disconnected reports and email threads to reviewed decisions with evidence. They must know which invoices are overdue, what customers actually promised, how cash maps to invoices, which posted transactions explain financial changes, when cash may breach policy, and which management commentary has been approved. A payment promise must never be mistaken for a receipt, and a model-generated explanation must never substitute for a reconciled accounting calculation.

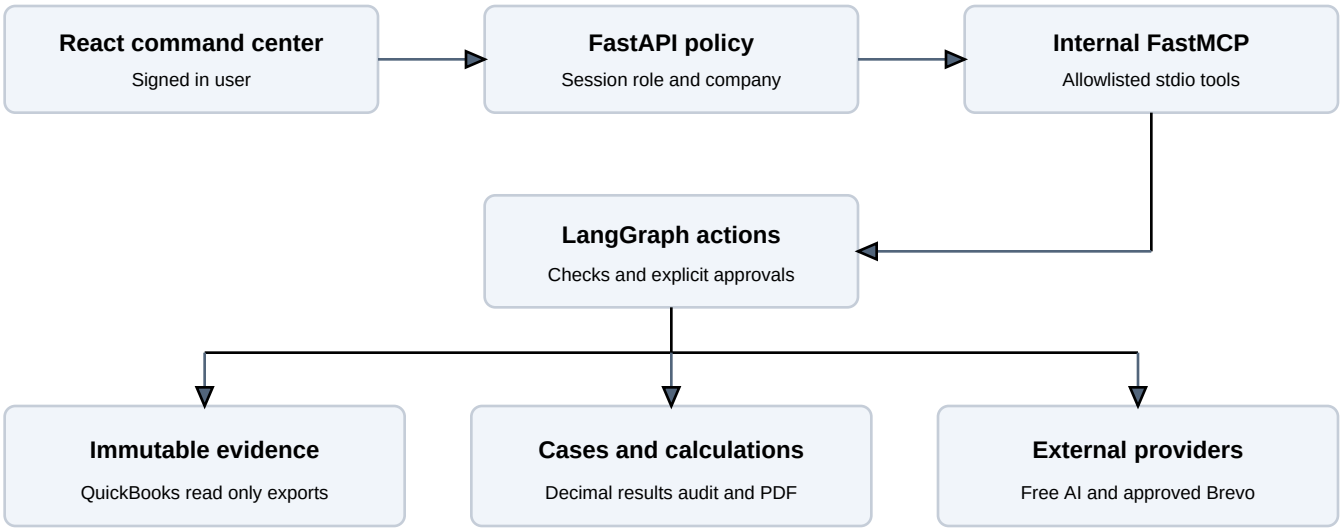
Scope and exclusions

- Included: React command center, authenticated accounts, server-side role and company controls, immutable source imports, five case journeys, LangGraph action orchestration, internal FastMCP tools, Brevo transactional delivery, reviewed reply import, calculated evidence, manual posting verification and retained PDF packs.

- Excluded: live installed QuickBooks Desktop SDK or Web Connector extraction, automated accounting mutations, bank transfers, payroll execution, automatic Brevo reply mailbox connection, guaranteed free-model availability, organizational SSO/MFA, certificate-based PDF signing and a managed production hosting SLA.

- Original `quickbooks_demo.sqlite` is an explicitly labelled fictional demonstration source. It does not become actual QuickBooks data by import, and missing transaction drivers, historical close evidence or bank reconciliation balances are not fabricated.

2 Architecture and trust boundaries



Accounting posting remains a Finance Admin action outside the application.

The browser never receives provider keys and cannot establish its own reviewer identity by changing a role field. FastAPI authenticates the cookie, resolves the principal and company scope, then stamps trusted reviewer email, user ID and an assigned role into MCP arguments. FastMCP runs as an internal stdio subprocess under the backend operator's authority. It is not exposed as a public remote MCP service. Direct shell access to that subprocess or the runtime database is therefore part of the trusted host boundary.

LangGraph is a governed action router rather than a background autonomous worker. Each request selects one allowlisted node, performs its checks, and returns to END. Human approvals occur in separate requests. State is persisted in SQLite with optimistic versions; there is no LangGraph checkpoint service or continuous scheduler. Source import and provider calls are explicit user-triggered operations.

Component responsibilities

Component	Implementation	Responsibility
UI shell and navigation	`frontend/src/main.jsx`	Role-filtered agents, workspace views, history Back/Forward and service health
Agent workspaces	`V11Workspace.jsx`, `AgentPortfolio.jsx`, `FinanceForms.jsx`	Case steps, evidence tables, chart, composer, review controls and manual tasks
Identity and policy	`authn.py`, `access.py`, `hosting.py`	Users, sessions, companies, trusted actor binding, origin checks and access audit
API	`v11_api.py`	Request bounds, routes, response filtering and MCP invocation
MCP boundary	`mcp_client.py`, `mcp_server.py`	Allowlisted business tool calls over stdio
Process orchestration	`v11_workflow.py`	Selection rules, state transitions, approval gates and closure

Component	Implementation	Responsibility
Import and provenance	`v11_source.py`, `sqlite_import.py`, `qbxml_import.py`	Validate and retain read-only source evidence
Finance engines	`v11_calculations.py`, `finance_models.py`, `v11_conversation.py`	Calculations and validated structured understanding
External delivery	`mail_gateway.py`, `brevo_gateway.py`	Prepare exact destination, verify sender and dispatch approved content
Persistence and packs	`v11_store.py`, `v11_manual.py`, `v11_reports.py`	Versions, audit, external posting verification and immutable PDF bytes

3 Source and data contracts

Client mode defaults to `FINANCE_CLIENT_MODE=1`. It permits `quickbooks_desktop_extract` and `quickbooks_enterprise_extract`. `quickbooks_demo_sqlite` requires explicit `FINANCE_ALLOW_SQLITE_DEMO=1`, which defaults to 0 for client installation. Illustrative fixtures and legacy sample routes are disabled in client mode. The active local demonstration may opt in to SQLite; the client package must not silently seed it.

Every normalized snapshot supplies `company_id`, `company_name`, `entity`, `currency`, `basis`, `as_of`, `source_type` and `source_reference`. Basis is Accrual or Cash; currency is a three-letter code. Source IDs, canonical JSON SHA-256 and server import timestamp are generated by the backend. Posted transaction account, period and date must agree with the mapped GL row; transaction and GL citations must point back to one another, and a transaction entity must match the snapshot. Equivalent content returns the existing immutable snapshot instead of overwriting it.

Collection	Important fields	Purpose
Customers and invoices	IDs, customer, number, dates, original amount, balance, terms, document reference	Collection eligibility and verified invoice facts
Payments	ID, customer, date, amount, bank reference, currency, invoice allocations	Duplicate checks and verification after external posting
Accounts and GL	Account classification, period, amount, report ID, transaction citations	Comparable P&L and account mapping
Transactions	ID, period/date, type, counterparty, account, signed amount/tax, Posted status, GL link, document	Reconciled drivers and correction proof
Bank receipts	Reference, payer/customer, date, amount, currency, cleared flag, proof reference	Cleared-cash allocation evidence

Collection	Important fields	Purpose
Contracts and plans	Invoice/customer references, terms/status, planned date/amount/category and approval	Supporting terms and approved cash outflows
Cash position	Same-date bank, QuickBooks and reconciliation balances, buffer, clearing flag, proof	Treasury opening cash gate
Reporting controls	Comparable Balance Sheets, close status, required checks, internal recipients	Management reporting review and distribution scope

Import bounds include JSON at approximately 5 MB, SQLite file at 10 MB, bank CSV at 2 MB, and bounded row collections. The SQLite reader uses a temporary copy with `mode=ro&immutable=1`, `query_only=ON`, `trusted_schema=OFF`, a fixed 25-table allowlist, a query progress budget and a maximum 10,000 rows per table. Virtual tables and unsupported provenance are rejected. It accepts one company per file. A real supported SQLite export declares `QUICKBOOKS_READONLY_EXPORT`; the original package declares `SYNTHETIC_DEMO`.

QBXML import covers Company, Customer, Invoice, ReceivePayment and Account query responses with completion checks. It does not synthesize missing P&L, Balance Sheet, close status or transaction drivers. To exercise all five agents, the client must provide the complete normalized financial and external evidence contract, not only an invoice export.

Persistence schema

The runtime SQLite database is a case and evidence store, not an accounting ledger. `v11_snapshots` retains immutable payloads and a unique digest. `v11_cases` retains agent, integer version and case JSON. `v11_tasks` adds a unique receipt/correction key and version. `v11_artifacts`, `v11_packs` and `v11_audit` retain document content, pack content and events. Authentication uses `auth_users`, `auth_sessions`, `auth_attempts` and `auth_events` in the runtime store.

A case contains its selected invoice/period/bank IDs, source snapshot/hash, status/version, original messages and replies, extraction results, commitments, disputes, review tasks, owned actions, calculated results, narrative, approvals and trace. Case saving uses `BEGIN IMMEDIATE` and `UPDATE ... WHERE version=expected`. A stale request fails instead of overwriting another reviewer's work. Pack bytes and the case binding are saved in one transaction.

4 Identity and permissions

`FINANCE_AUTH_MODE=users` is the default. There is no public registration. A bootstrap operator creates the first owner, who must replace the temporary password before accessing finance APIs. Administrators create users and assign roles and company IDs; * is explicit all-company access. Password hashing uses PBKDF2-HMAC-SHA256, random salt and 600,000 iterations. Sessions use random opaque tokens whose SHA-256 digests are retained, eight-hour expiry and user-version binding. Cookies are HttpOnly and SameSite Strict, with Secure enabled for hosted HTTPS.

Role	Agent access	Permitted decisions	Explicit restriction
AR Reviewer	AR	Messages, supporting documents and promises	No accounting posting or financial report sign-off
Cash Analyst	Bank	Payer/allocation review and task preparation	Cannot approve, record or verify manual posting task
Finance Admin	AR and Bank	Source imports and manual task approval, external ID recording, fresh-source verification	Cannot approve FP&A, treasury or CFO gates
Finance Head	FP&A	Trends, drivers, commentary, actions and approved pack export	No customer collections email or CFO authority
Treasury Head	Treasury	Schedules, probability assumptions, scenarios and mitigations	No accounting write or CFO sign-off
Controller	Reporting	Evidence resolution and current pack review	No final CFO distribution authority
CFO	Treasury and Reporting	Independent final approval, approved export and internal distribution	Cannot approve their own preceding Controller or Treasury Head review
System Admin	No finance agent by this role alone	Users, roles, company scope, resets and security events	Financial decisions require a separately assigned finance role

Company scope applies to cases, tasks, snapshots, evidence files, artifacts and packs. Source responses remove collections and raw SQLite tables outside the user's finance role. Finance Admin can see relevant posting tasks without receiving the full FP&A case. Browser-supplied reviewer text is replaced with the authenticated email and ID. The server rejects unassigned role claims. A new CFO gate requires a different authenticated user ID from the prior Controller/Treasury Head reviewer; historical free-text reviewer labels do not satisfy it.

Login throttling applies after 20 account/IP attempts in 15 minutes. Logout, password changes/resets, deactivation and role/company changes revoke affected sessions. The last active System Admin cannot be removed. Global provider administration requires an all-company System Admin. Hosted mutations require an allowed origin; localhost requests are also authenticated in users mode. SSO/MFA and enterprise identity lifecycle integration remain client production choices.

5 Shared message and evidence controls

Message states are `draft to approved to sending to sent`; rejected dispatch becomes `send_failed`, and uncertain network/provider outcomes become `send_unknown`. Editing subject, body, recipient or attachments returns the message to draft and removes approval. The message hash binds subject, body, recipient, artifact IDs, purpose and pack ID. Artifact digests and the prepared delivery provider/sender/destination are retained in approval. Actual QuickBooks customers must use the verified imported contact; demo recipients can be edited. Internal reporting recipients must come from the explicitly configured list.

Brevo preparation checks its configured sending enable flag, API credential and active verified sender before approval. The saved approved address is used exactly; it is not silently changed by the demo default at dispatch. The UI defaults to provider delivery, with a separate explicit local-outbox simulation option for labelled demonstrations. Changing the checkbox after approval does not rewrite the saved approval; reapproval is required to change an already approved transport.

Sending is a separate explicit action after recipient, mapping, attachment and tone checks. The case version is atomically claimed before dispatch. The idempotency key derives from case ID, message ID and approval hash. A sent message is not resent by a repeated action. A timeout or missing provider message ID is not treated as failure safe to retry; the user must reconcile the provider log first. Brevo acceptance means queued by the provider, not proved delivered to the inbox.

Replies in the Brevo path are imported with original body, sender, reply identity and In-Reply-To evidence. Brevo sending does not automatically provision an inbound mailbox. Sender and thread checks precede extraction. Reimporting one message ID with altered content is rejected; later replies retain prior original messages. Automatic provider fetch applies only to separately authorized Gmail/Resend adapters. Provider connection checks never send mail.

Artifacts are bounded, content-hashed and invoice-scoped. They require approval before outbound attachment and must match the case company/invoice selection. Supporting file access uses retained references and allowlists, not arbitrary URLs or server paths. A requested timesheet cannot be satisfied by an unrelated approved document.

6 AR Collections process

Slides 3-8 define a customer and invoice conversation loop. The portfolio derives open AR, aging buckets, priority customer balances, approved promise totals and unresolved review counts from source rows and cases. Any invoice past its due date with a positive open balance is eligible for an AR case. The greater-than-30-day measure remains a priority metric and does not exclude younger overdue invoices from collection initiation.

1. Select company/source, customer and eligible open invoices. Retain the selected source hash.

2. Draft verified invoice numbers, balances, dates and terms with optional approved documents. AI can draft subject/closing but cannot replace verified facts.

3. Save edits, complete the recipient/mapping/tone checks, approve the exact message and explicitly send.

4. Import and verify the original reply. Extract invoice-level tranches, disputes, artifact requests, missing dates and quoted evidence.

5. Resolve unknown dates and disputed instructions by a clarified customer reply, not a finance note inventing a date. Approve invoice-matching documents separately.

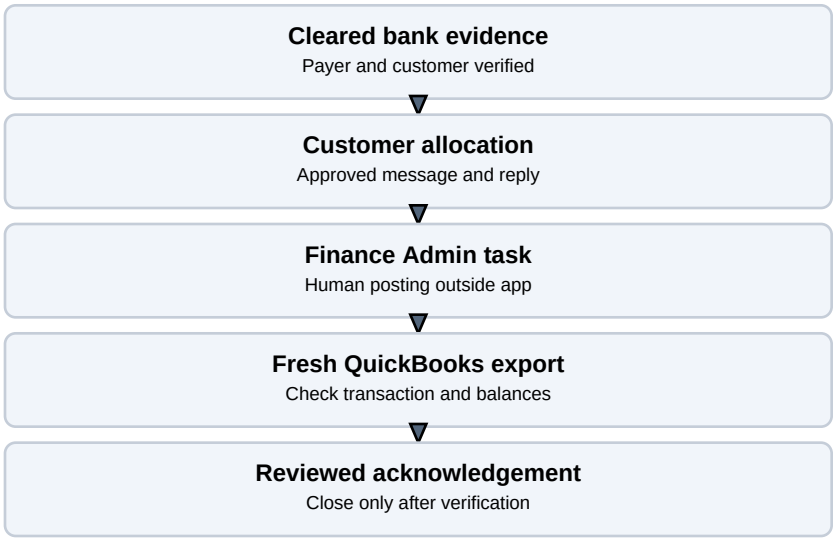
6. Approve the supported schedule, retain it in the linked case store and monitor follow-up actions. A promise does not change invoice balance or create a QuickBooks payment. Cleared actual cash enters the bank workflow.

The extraction schema is strict JSON. Amounts must be positive, finite Decimal-compatible values. Each tranche identifies a known invoice, amount, explicit date or null, verbatim quote, inference flag and confidence. Validation checks invoice and amount support in the original quote, quoted date support, duplicates, confidence below 0.8, totals against open balances and conditional language. Partial amounts plus disputes may not exceed the invoice balance. “Now” or “initially” does not establish an invented calendar date. Schedule approval requires resolved document/ambiguity tasks, no disputes and current dates/balances.

Slide acceptance examples include both invoices on November 12, an approved timesheet request, $6,000 initially plus $9,000 November 20 with first date unresolved, and $4,000 approved plus $6,000 disputed. These are test fixtures, not client accounting records.

7 Bank to invoice reconciliation process

Slides 9-12 require cleared bank evidence and customer instructions before posting preparation. A bank amount alone cannot identify a payer or invoice allocation.



Allocation arithmetic uses `sum(invoice allocations) = cleared receipt amount`. Every allocation must be positive, within its invoice's open balance, unique by invoice and supported by explicit reply evidence. The payer, currency and reference must match; duplicate bank references or an existing payment block a new task. A unique task key hashes company, entity and reference.

Manual task states are `proposed to approved to recorded_unverified to verified`. Recording a transaction ID is not verification. Verification needs a different digest and an import timestamp after the external posting was recorded, the same company/entity/currency/basis/source type, one matching payment ID/reference, correct payment date, customer, amount/currency, allocations and exact remaining invoice balances. Acknowledgement is blocked until verification. The app never performs Receive Payment itself.

8 FP and A Trend Intelligence process

Slides 13-20 describe monthly financial analysis. Select a matching entity/basis source and date range; the engine requires at least two comparable, consecutive monthly periods and uses six months when present. Explicit selected start/end months must be present; missing or skipped periods are rejected. Missing months and transaction evidence are not invented. Each month requires explicit Income, CostOfGoodsSold and Expense evidence, including a zero row where a category truly has no activity; an absent category creates a blocking issue rather than silently establishing zero. The user can inspect revenue, cost, expense and margin graphs and drill to source transactions. FP&A has no customer email process.

For each month:

- Revenue = mapped Income; cost of services = CostOfGoodsSold; operating expense = Expense.

- Gross profit = revenue - cost of services.

- Total operating cost = cost of services + operating expense.

- Operating profit = revenue - cost of services - operating expense.

- Operating margin percentage = 100 × operating profit / revenue; zero revenue gives undefined, not division by zero.

- MoM change percentage = 100 × (current - prior) / prior; zero baseline is undefined.

- Profit change = revenue change - cost of services change - operating expense change.

Account drivers use only Posted transactions. Tax-inclusive transaction amounts are normalized net of tax; signed credit memos/refunds remain signed. OtherIncome and OtherExpense remain distinct non-operating account classifications and are not silently merged into operating revenue or cost. Their presence is disclosed in source caveats; this operating analysis is not a full net-income calculation. Prior-period transaction amounts are subtracted from current-period amounts and reconciled with the GL account delta. A nonzero unexplained delta blocks approval. Contributor records retain customer/vendor, transaction IDs, document references, GL IDs and report periods. An absolute revenue MoM change of at least 10% is the implemented descriptive outlier flag. Year-over-year comparison exists only for sourced prior-year months. Seasonal indices need at least 24 selected observations and two occurrences of the calendar month; they are descriptive rather than predictive forecasting.

In the six-month deck fixture, September revenue 540,000 minus costs 318,000 and operating expense 137,000 gives profit 85,000 and margin approximately 15.7407%. August profit is 67,000 on revenue 480,000. The exact margin movement is approximately 1.7824 percentage points; subtracting the separately rounded displayed margins gives 1.7 points. Both meanings are distinguished. Slide 31's 9.8% expense example does not reconcile to the supplied numbers: operating expense growth is 9.6%, while total cost growth is approximately 10.17%. Source arithmetic takes precedence over copying the mockup value.

Narrative generation receives deterministic fact IDs and source references. Every highlight must cite known facts; model prose cannot contain digits, dollar signs or percent signs that could introduce unsupported numeric assertions. Business explanations require finance review and supporting evidence. Finance Head signs off calculated results and commentary, owns follow-up actions and exports the current approved pack.

Manual correction proposals require current-period GL evidence, substantive rationale, mapped income/cost/expense accounts and balanced debit/credit lines. Finance Admin posts outside the application. A new source must prove the exact posted journal account impacts and the resulting GL change before recalculation and review can continue.

9 Treasury forecast process

Slides 21-28 require bank-verified cash, approved invoice schedules, explicit timing assumptions and a 13-week view. Opening cash must be cleared, dated to the snapshot and reconcile with both the QuickBooks cash balance and an explicitly supplied QuickBooks reconciliation balance. A missing reconciliation balance is rejected rather than defaulted from cash. The engine refuses to infer a QuickBooks balance from a bank-only source.

Treasury Head reviews imported AR commitments again, including their correspondence, invoice/MSA references, disputes and document requirements. Imported schedules exceeding refreshed invoice balances are rejected. Each included tranche must have a date and approval. Optional receipt probabilities are between zero and one and need a rationale when changed from certainty.

For each approved receipt, expected amount = nominal amount × approved probability. For each day over 91 days, cash closes as prior cash + expected receipts - approved outflows. Headroom = cash - policy buffer; shortfall = max(0, buffer - cash). Thirteen seven-day summaries expose inflow, outflow, ending cash, minimum cash and maximum shortfall. The forecast also identifies first breach and maximum shortfall across the full

horizon.

Base preserves approved event timing. Optimistic and Stress shift portions of existing events to explicit dates, with amount bounds and a substantive note. Scenario changes do not create money or accounting transactions. In the deck fixture, 250,000 opening +150,000 receipts -300,000 outflows =100,000 ending cash; buffer 125,000 gives ending shortfall 25,000. Optimistic 155,000 and Stress 45,000 require explicit 55,000 event shifts across the horizon boundary; merely changing dates within the same horizon does not change the total.

Treasury Head approves current assumptions and results; a distinct CFO gives final cash-plan approval. A buffer breach anywhere within the horizon, including an intermediate shortfall that recovers before the final day, requires an owned mitigation action before CFO approval. Owned mitigation actions record responsible person, due date and outcome. Actual receipts enter bank verification rather than being booked from promises. No payment rescheduling, funding or vendor execution is performed by the app.

10 Financial Reporting process

Slides 29-31 require comparable P&L, Balance Sheet, aging, GL and close evidence. The engine requires the explicitly selected current report month to be sourced and compares it with the immediately preceding calendar month, calculates the financial trend pack, checks exactly one Balance Sheet per period and verifies assets = liabilities + equity within the half-cent tolerance. A period must be Closed. Unexplained transaction-to-GL movements remain blocking differences.

Aging retains the actual snapshot as-of date. Current invoice balances must not be labelled historical month-end balances. If dates differ, the pack raises a scope task that the Controller must resolve using evidence or explicitly retain as a disclosed limitation. Financial findings cannot be approved with unresolved required checks. Unreconciled accounting differences require corrected source evidence, not a cosmetic note.

Controller approval binds the current source, selection, result, narrative, commitments, review tasks and actions. A different authenticated CFO approves the same content. PDF export requires a current approval fingerprint and stores immutable PDF bytes, source hash, content hash, reviewer records and timestamps. Pack retrieval checks byte integrity. These application approvals are not a cryptographic certificate signature.

CFO distribution permits only explicitly configured internal recipients. A separately approved exact message carries the exact saved PDF. Editing reviewed figures, assumptions, checks, narrative or actions invalidates the current pack binding and requires fresh approval; previous PDF records remain retained. Completed/distributed cases are immutable and require a new case for revisions.

11 AI provider behavior and failure handling

The active free design uses the saved backend OpenRouter key with `openrouter/free` or a `:free` catalogue model. Before generation the code checks the current catalogue entry and zero prompt/completion price. It requests bounded JSON with required structured-output parameters. Returned content must parse as an object and pass each workflow schema and evidence validator. Missing content, truncated output, invalid fields or unsupported quotes stop the action; deterministic rules are used only when the provider mode is explicitly demo.

Transient OpenRouter 429/503 responses permit one retry of the model generation only. A numeric or HTTP-date Retry-After is respected when the wait is at most five seconds; a longer wait is reported without blocking the request. Daily free quota and account/key exhaustion do not cause retry storms or a paid fallback. Provider/model/generation provenance is retained for generated email drafts, narrative and extraction results.

Template-only messages are identified as non-live source templates.

Direct Claude remains an optional configured adapter. Successful key authentication does not prove generation quota. Insufficient API credits, rejected credentials and usage limits have distinct user messages. A failed direct Claude request does not silently charge another provider. Selection of OpenRouter is an explicit configuration choice. Free service availability and quotas are external constraints and cannot be guaranteed for client production.

12 API and MCP contract

Operation	API	MCP tool or handler
Session and login	`/api/auth/session`, `/login`, `/logout`, `/change-password`	Authentication handlers
Accounts and access	`/api/auth/users`, `/security-audit`, `/access-context`	Administrator policy handlers
Workspace and source	`/api/v11/workspace`, `/snapshots/{id}`, `/export-guide`	Filtered read handlers
Import normalized JSON	POST `/api/v11/snapshots`	`v11_import_snapshot`
Import SQLite	POST `/api/v11/import-sqlite`	`v11_import_sqlite`
Import QBXML	POST `/api/v11/import-qbxml`	Normalize, then `v11_import_snapshot`
Import statement	POST `/api/v11/snapshots/{id}/bank-statement`	`v11_import_bank_statement`
Create/action case	POST `/api/v11/cases`, `/cases/{id}/actions`	`v11_open_case`, `v11_case_action`
Posting task action	POST `/api/v11/tasks/{id}/actions`	`v11_manual_action`
Artifact upload/approval	`/api/v11/cases/{id}/artifacts`, `/artifacts/{id}/approve`	`v11_upload_artifact`, `v11_approve_artifact`
Save reviewed pack	POST `/api/v11/cases/{id}/save-pack`	`v11_save_pack`
Evidence and pack retrieval	`/snapshots/{id}/evidence-file`, `/cases/{id}/evidence.csv`, `/packs/{id}.pdf`	Scoped read handlers
Provider readiness	`/api/v11/email-status`, POST `/check-brevo`	`brevo_connection_status` for read-only verification

Mutating case requests include `action`, positive `expected_version` and bounded payload. Unknown top-level request fields are rejected where modeled. Business validation errors are surfaced to the user; stale versions require refresh. Public health exposes readiness policy but not credentials. Secrets remain

backend-only; configuration rejects secret-looking values in model/display fields, and blank secret updates preserve saved credentials.

13 Test driven validation strategy

The regression suite uses isolated temporary runtime databases and explicit demo fixtures. Those fixtures verify arithmetic and transitions without being displayed as client accounting. Legacy-test auth mode is isolated to legacy tests; dedicated user-access tests exercise users mode and actual server policy. No test should send real mail, modify QuickBooks or invoke bank operations.

Acceptance area	Required checks	Test source
Read-only integrity	Original database hash unchanged; no accounting mutation path	`test_v11.py`, `test_sqlite_import.py`, `test_service_integrations.py`
AR understanding	Four deck paths, unknown dates/IDs, disputes, supporting documents and clarified schedule retention	`test_v11.py`
Bank/manual lifecycle	Totals, duplicate/uncleared guards, real transaction ID, new-source proof, balance reconciliation and acknowledgement	`test_v11.py`
Financial algorithms	Trend arithmetic, transaction reconciliation, exact profit bridge, manual journal balance and impact	`test_v11.py`
Treasury	Three tranches, 91 days/13 weeks, probability review, explicit scenario shifts and cash evidence	`test_v11.py`
Reporting	Controller/CFO checks, immutable PDF, close and source scope	`test_v11.py`, `test_user_access.py`
Identity isolation	Anonymous denial, role spoofing, company/evidence isolation, independent CFO and session revocation	`test_user_access.py`
Provider errors	Key-field protection, insufficient Claude credits, bounded OpenRouter retry and quota classification	`test_provider_diagnostics.py`
Exact Brevo recipient	Edit resets approval, exact destination reaches mocked provider, repeated click does not duplicate	`test_v11.py`
Client packaging	No secrets/runtime records/personal hosting bindings; empty client boot and protected APIs	`scripts/verify_client_package.py`

Release verification must run the complete backend suite, production frontend build, source/hash checks and sanitized package verification against the final source. UI review must use representative accounts for all roles,

every agent step, recipient editing, required checklists, Back/Forward and unauthorized destinations. Automated backend tests do not prove browser usability or actual provider delivery. Live sender readiness checks are read-only. A client-approved test email should separately verify provider acceptance, delivery event and reply ingestion before customer use.

14 Deployment and operations

The current free preview pattern is a Vercel static React frontend whose `/api` rewrites proxy through an HTTPS tunnel to a backend on the operator's computer. Local development uses Vite on 5174 and FastAPI on 8000. The hosted wrapper binds a protected backend on 8001 and loads hosted settings from a private file. Frontend builds contain neither provider credentials nor user passwords. Temporary tunnel names and process uptime are not durable client infrastructure.

The client installation must own its accounts, backend host, HTTPS origin configuration, verified email sender and authorized accounting export pipeline. Persist the runtime database on durable storage with restricted permissions. Backups must include case/evidence/auth data and any required local OAuth encryption key without shipping secrets in source archives. Test restoration and session revocation on a restored database. Record source timestamps and hashes so exported evidence can be traced back to the client's extraction process.

Failures require distinct handling: a model rate limit asks for controlled retry; invalid evidence asks for corrected source; a case conflict asks for refresh; rejected email retains a failed status; uncertain delivery requires provider reconciliation; unverified posting keeps accounting tasks open. Operators must not clear these controls solely to make a demonstration proceed.

15 Client delivery gates and acceptance plan

1. Confirm the agreed scope is read-only exports and manual accounting, and approve the handling of deck arithmetic inconsistencies.

2. Provide a complete authorized client source: six financial periods where required, posted transaction links, comparable Balance Sheets, close status, bank cash reconciliation, contracts and approved outflows. The small original SQLite package is insufficient for all financial sign-offs by itself.

3. Complete owner first login, assign least-privilege users/company scopes and establish a distinct Controller/Treasury Head and CFO for final approval. Remove temporary credentials from handover paths after enrollment.

4. Configure client-owned provider keys and verified Brevo sender. Rotate any credential exposed during earlier setup. Run an approved send/delivery/reply acceptance scenario using a controlled recipient.

5. Complete every five-agent acceptance journey on client-authorized evidence, including rejection paths and manual posting verification against a genuinely refreshed export.

6. Review final code findings and resolve release-blocking logic/control defects. Verify the source-to-slide coverage matrix for all 32 physical slides; do not mark UI mockup metrics as actual client balances.

7. Confirm production hosting, backup/restore, monitoring, retention, support owner, incident procedures and external AI data-sharing approval. Free preview services remain suitable only where their quotas and uptime match the agreed use.

8. Generate the final sanitized client package and fresh verification record. Have the finance owner and technical owner sign the acceptance record with any explicitly retained limitations.

Role interpretation to accept

Slide 21 names Finance Head approval of mitigations; slides 27 and 28 use Treasury Head and CFO. The implementation currently assigns scenario and mitigation review to Treasury Head and requires an independent CFO final decision. Finance Head is scoped to FP&A. The finance owner must approve this mapping or request a separate Finance Head mitigation gate before production acceptance.

16 Source references and document maintenance

Primary requirements: `Finance and Accounting AI agentsV1.1.pptx`; readable extraction: `docs/17-v1.1-slide-transcript.md`. Current implementation sources are the component files in Section 2. `docs/24-authentication-and-access-control.md` expands identity controls; `docs/23-sqlite-and-brevo.md` records source/provider setup. Historical documents may describe earlier outbox-only, reviewer-assertion or shared-password behavior and must not supersede the current source.

Update this TDD whenever a schema, action, role, source policy, provider transport or approval hash changes. The release record must name the reviewed version, fresh tests, actual deployment artifact and unresolved acceptance gates. An architecture diagram or passing fixture test alone is not client acceptance.